

Document filename: 290426ak.sqx
Spectrum ID:
Model ID: Default Model

EXPERIMENTAL CONDITIONS

Excitation mode: X-ray fluorescence (XRF)
Sample detector path: Air Detector type: Si(Li)
Path length: 3.0 cm Dead layer: 0.0300 µm
Primary energy: 16.5 Kev Contact layer: 0.0200 µm
Incidence angle: 45.0 degrees Active depth: 3.0 mm
Take-off angle: 45.0 degrees Window: 760.00 µm
Pulse pile-up resolution time: 2.00 µs

Fitting Region: channels 91 - 1634, 4 iterations done, last change = 0.09%

Lambda = 1e-006 , ChiSquare = 3.8

CALIBRATION DATA

	Initial estimate	Final estimate
ZERO (eV)	-387.2 ± 100.0	-391.6 ± 0.5
GAIN (eV/ch)	14.673 ± 2.000	14.649 ± 0.002
NOISE (eV)	81.5 ± 40.0	56.7 ± 2.8
FANO factor	0.155 ± 0.050	0.139 ± 0.003

PEAK DATA

# Line	E(KeV) chan#	Rel. int. fwhm (eV)	Peak area backgr	st. dev	Chi-sq tot. abs
1 Si K			413 ± 45		
KA1	1.740	1.00000	413 ± 45		3.4
	145.510	91.58	836		3.63e-008
2 P K			747 ± 57		
KA1	2.013	1.00000	747 ± 57		1.8
	164.146	95.92	1224		1.49e-005
3 S K			3295 ± 127		
KA1	2.307	0.80932	2667 ± 77		2.3
	184.216	100.37	1855		5.81e-004
KB1	2.464	0.19068	628 ± 50		11.8
	194.933	102.68	2232		2.22e-003
4 Cl K			37368 ± 270		
KA1	2.622	0.82302	30755 ± 183		32.3
	205.719	104.94	3000		6.32e-003
KB1	2.816	0.17464	6526 ± 69		60.3
	218.962	107.66	3333		1.66e-002
KB1 Escape	1.074	0.00235	88 ± 18		5.4
	100.045	80.03	341		7.09e-027
5 Ar K			15590 ± 358		
KA1	2.957	0.81836	12758 ± 238		27.3
	228.588	109.59	3475		2.91e-002
KB1	3.191	0.16920	2638 ± 76		29.2
	244.562	112.72	3516		5.90e-002
KA1 Escape	1.215	0.01046	163 ± 20		3.3
	109.671	82.61	380		1.92e-022
KB1 Escape	1.449	0.00199	31 ± 24		3.7
	125.645	86.73	580		1.45e-013
6 K K			83958 ± 433		
KA1	3.313	0.82516	69279 ± 296		14.5
	252.890	114.32	3465		7.89e-002
KB1	3.590	0.16393	13763 ± 81		25.6
	271.799	117.87	3201		1.36e-001
KA1 Escape	1.571	0.00925	777 ± 26		3.9
	133.973	88.80	687		8.15e-011
KB1 Escape	1.848	0.00165	139 ± 30		12.0
	152.882	93.32	926		5.92e-007
7 Ca K			22436 ± 313		
KA1	3.691	0.83279	18685 ± 176		23.7
	278.694	119.14	3072		1.59e-001
KB1	4.013	0.15777	3540 ± 63		2.5
	300.675	123.09	2999		2.40e-001
KA1 Escape	1.949	0.00809	181 ± 33		9.2
	159.777	94.92	1085		4.84e-006
KB1 Escape	2.271	0.00135	30 ± 42		3.7
	181.758	99.84	1735		4.18e-004
8 Ti K			806 ± 251		
KA1	4.509	0.85258	687 ± 77		1.1
	334.534	128.95	2841		3.66e-001

KB1	4.932	0.14056	113	± 58	1.0
	363.410	133.74	3228		4.65e-001
KA1 Escape	2.767	0.00602	5	± 57	53.3
	215.617	106.98	3270		1.33e-002
KB1 Escape	3.190	0.00084	1	± 59	29.2
	244.493	112.71	3516		5.89e-002
9 Cr K			600	± 265	
KA1	5.412	0.86647	520	± 89	1.9
	396.177	138.98	4105		5.60e-001
KB1	5.947	0.12866	77	± 68	1.6
	432.699	144.60	4545		6.47e-001
KA1 Escape	3.670	0.00434	3	± 56	22.1
	277.260	118.88	3092		1.54e-001
KB1 Escape	4.205	0.00053	0	± 52	1.2
	313.782	125.39	2717		2.89e-001
10 Mn K			3013	± 286	
KA1	5.895	0.86610	2609	± 107	2.0
	429.149	144.06	4513		6.39e-001
KB1	6.491	0.12983	391	± 71	1.8
	469.835	150.09	4861		7.15e-001
KA1 Escape	4.153	0.00363	11	± 53	0.9
	310.232	124.78	2768		2.76e-001
KB1 Escape	4.749	0.00043	1	± 55	1.9
	350.918	131.69	3055		4.24e-001
11 Fe K			41612	± 408	
KA1	6.399	0.86901	36161	± 213	1.3
	463.554	149.18	4820		7.05e-001
KB1	7.059	0.12764	5311	± 80	2.3
	508.609	155.62	5603		7.71e-001
KA1 Escape	4.657	0.00300	125	± 55	1.6
	344.638	130.65	2975		4.02e-001
KB1 Escape	5.317	0.00035	14	± 60	2.0
	389.692	137.96	3563		5.43e-001
12 Co K			1174	± 312	
KA1	6.925	0.87101	1023	± 110	1.4
	499.462	154.33	5014		7.59e-001
KB1	7.649	0.12623	148	± 76	1.1
	548.885	161.16	5658		8.15e-001
KA1 Escape	5.183	0.00248	3	± 59	0.9
	380.545	136.51	3449		5.17e-001
KB1 Escape	5.907	0.00028	0	± 67	2.0
	429.968	144.18	4513		6.41e-001
13 Ni K			1097	± 447	
KA1	7.478	0.57714	633	± 90	1.2
	537.212	159.58	5656		8.04e-001
KA2	7.461	0.29531	324	± 79	1.5
	536.051	159.42	5655		8.02e-001
KB1	8.265	0.12527	137	± 76	2.8
	590.936	166.75	5595		8.51e-001
KA1 Escape	5.736	0.00135	1	± 66	2.7
	418.295	142.41	4387		6.15e-001
KA2 Escape	5.719	0.00070	1	± 66	2.7
	417.134	142.23	4375		6.12e-001
KB1 Escape	6.523	0.00023	0	± 70	2.0
	472.019	150.41	4881		7.19e-001
14 Zn K			718	± 460	
KA1	8.639	0.57753	415	± 89	1.1
	616.467	170.06	5511		8.68e-001
KA2	8.616	0.29663	213	± 78	1.2
	614.897	169.86	5518		8.67e-001
KB1	9.572	0.12429	89	± 76	0.9
	680.158	178.03	5727		9.01e-001
KA1 Escape	6.897	0.00093	1	± 71	1.4
	497.550	154.06	5006		7.57e-001
KA2 Escape	6.874	0.00048	0	± 71	1.2
	495.980	153.84	4998		7.54e-001
KB1 Escape	7.830	0.00015	0	± 75	2.8
	561.241	162.83	5650		8.27e-001
15 Rb K			738	± 476	
KA1	13.395	0.55983	413	± 79	1.4
	941.134	207.54	4424		9.58e-001
KA2	13.336	0.29068	215	± 70	1.0
	937.106	207.12	4459		9.57e-001
KB1	14.958	0.13370	99	± 60	1.2
	1047.831		218.46	3527	9.67e-001
KB2	15.185	0.01534	11	± 58	1.1
	1063.327		220.00	3405	9.69e-001
KA1 Escape	11.653	0.00026	0	± 71	2.8
	822.217	194.65	5099		9.44e-001
KA2 Escape	11.594	0.00014	0	± 72	2.6
	818.189	194.20	5134		9.43e-001
KB1 Escape	13.216	0.00005	0	± 65	1.3
	928.914	206.26	4187		9.56e-001
16 Ag K			397	± 168	
KA1	22.163	0.65194	259	± 46	1.1

	1539.677	263.00	1296	8.79e-001
KA2	21.990 0.34795	138	± 40 1.1	
	1527.868	262.02	1347	8.85e-001
KA1 Escape	20.421 0.00007	0	± 41 4.3	
	1420.760	252.95	1673	9.26e-001
KA2 Escape	20.248 0.00004	0	± 41 3.1	
	1408.951	251.93	1717	9.30e-001

17Ag L		35156	± 1095	
L1M3	3.234 0.11071	3892	± 77 18.1	
	247.497 113.29	3501	6.51e-002	
L1M2	3.203 0.06168	2168	± 65 23.3	
	245.381 112.88	3512	6.01e-002	
L1N3	3.747 0.08968	3153	± 68 36.7	
	282.517 119.84	2992	1.73e-001	
L1M5	3.437 0.00275	97	± 58 8.6	
	261.355 115.92	3363	1.02e-001	
L2M4	3.151 0.23201	8156	± 120 48.8	
	241.831 112.19	3524	5.29e-002	
L2N4	3.520 0.05488	1929	± 62 22.7	
	267.021 116.98	3269	1.20e-001	
L2M1	2.806 0.00206	73	± 58 60.3	
	218.280 107.52	3333	1.59e-002	
L2N1	3.428 0.00251	88	± 58 6.1	
	260.740 115.81	3377	1.01e-001	
L3M5	2.984 0.33280	11700	± 161 32.4	
	230.431 109.95	3492	3.21e-002	
L3N5	3.348 0.09173	3225	± 72 5.7	
	255.279 114.77	3437	8.54e-002	
L3M1	2.634 0.00260	91	± 55 31.7	
	206.538 105.11	3031	6.77e-003	
L3N1	3.256 0.00516	181	± 59 16.9	
	248.999 113.58	3495	6.88e-002	
L1M3 Escape	1.492 0.00128	45	± 25 2.2	
	128.580 87.46	620	1.71e-012	
L1M2 Escape	1.461 0.00073	26	± 24 3.7	
	126.464 86.93	593	2.97e-013	
L1N3 Escape	2.005 0.00085	30	± 35 3.1	
	163.600 95.79	1197	1.30e-005	
L2M4 Escape	1.409 0.00276	97	± 21 5.5	
	122.914 86.04	452	1.10e-014	
L2N4 Escape	1.778 0.00057	20	± 29 6.2	
	148.104 92.20	869	1.06e-007	
L3M5 Escape	1.242 0.00421	148	± 20 3.8	
	111.514 83.10	388	4.60e-021	
L3N5 Escape	1.606 0.00102	36	± 27 5.5	
	136.362 89.38	726	3.57e-010	

18Gd L		2122	± 2037	
L1M3	6.831 0.08253	175	± 72 1.2	
	493.045 153.42	4990	7.50e-001	
L1M2	6.687 0.05619	119	± 71 2.6	
	483.215 152.02	4943	7.36e-001	
L1N3	8.105 0.02330	49	± 75 3.9	
	580.014 165.32	5620	8.42e-001	
L1N2	8.087 0.01564	33	± 75 4.7	
	578.785 165.16	5624	8.41e-001	
L1O3	8.355 0.00547	12	± 75 2.3	
	597.080 167.55	5575	8.55e-001	
L1M5	7.192 0.00192	4	± 75 5.3	
	517.688 156.89	5628	7.82e-001	
L1M4	7.160 0.00129	3	± 75 4.9	
	515.504 156.58	5623	7.79e-001	
L2M4	6.713 0.27516	584	± 80 2.6	
	484.990 152.28	4948	7.39e-001	
L2N4	7.786 0.05649	120	± 76 1.8	
	558.237 162.42	5653	8.24e-001	
L2M1	6.050 0.00675	14	± 68 3.8	
	439.730 145.65	4616	6.61e-001	
L2N1	7.554 0.00198	4	± 75 0.9	
	542.400 160.28	5658	8.09e-001	
L2O1	7.894 0.00034	1	± 75 4.0	
	565.610 163.41	5646	8.31e-001	
L3M5	6.057 0.33572	712	± 82 4.5	
	440.208 145.72	4626	6.62e-001	
L3M4	6.025 0.03779	80	± 68 3.3	
	438.023 145.40	4606	6.58e-001	
L3N5	7.103 0.07842	166	± 76 4.0	
	511.613 156.04	5612	7.75e-001	
L3M1	5.362 0.01292	27	± 64 2.0	
	392.764 138.45	4050	5.51e-001	
L3N1	6.867 0.00414	9	± 71 1.2	
	495.502 153.77	4998	7.54e-001	
L3O1	7.207 0.00071	2	± 75 5.5	
	518.712 157.03	5630	7.83e-001	
L1M3 Escape	5.089 0.00024	1	± 58 0.8	
	374.128 135.48	3372	4.98e-001	
L1M2 Escape	4.945 0.00017	0	± 57 0.8	
	364.298 133.89	3241	4.67e-001	
L1N3 Escape	6.363 0.00004	0	± 69 3.2	
	461.097 148.82	4806	7.01e-001	
L1N2 Escape	6.345 0.00003	0	± 69 4.5	
	459.868 148.64	4791	6.99e-001	

L2M4 Escape	4.971	0.00085	2	± 57	0.9
	366.073	134.18	3267		4.73e-001
L2N4 Escape	6.044	0.00012	0	± 68	3.8
	439.320	145.59	4616		6.60e-001
L2M1 Escape	4.308	0.00003	0	± 52	1.1
	320.813	126.61	2679		3.16e-001
L3M5 Escape	4.315	0.00132	3	± 52	1.1
	321.291	126.69	2684		3.18e-001
L3M4 Escape	4.283	0.00015	0	± 52	1.1
	319.107	126.32	2678		3.09e-001
L3N5 Escape	5.361	0.00021	0	± 64	2.0
	392.696	138.43	4050		5.51e-001
L3M1 Escape	3.620	0.00007	0	± 56	28.8
	273.847	118.25	3166		1.43e-001
L3N1 Escape	5.125	0.00001	0	± 58	0.7
	376.585	135.87	3398		5.05e-001

Filter Background
